Shenran Wang

shenranw@cs.ubc.ca | Google Scholar | GitHub

Research Interests

My research interests spin around efficiency and robustness. I enjoy defining open-ended problems and delivering elegant solutions that lie on the sweet spot between utility, reliability and efficiency.

Education

The University of British Columbia
M.Sc. in Computer Science 2025.9 – 2027.5
Advisors: Vered Shwartz and Hila Gonen, Natural Language Processing Lab
B.Sc. in Computer Science 2020.9 – 2025.5
Research advisor: Jian Zhu, Natural Language Processing Lab

Publications

Shenran Wang, Vered Shwartz, and Hila Gonen. *Don't 'Well, Actually' Me Unless You Know What You're Talking About: Weak Presupposition Verification Degrades General QA Performance*. arXiv:2608.06539, 2026; under review.

Shenran Wang, Timothy Tin-Long Tse, and Jian Zhu. *Understanding In-Context Learning Beyond Transformers: An Investigation of State Space and Hybrid Architectures.* Findings of ACL 2026.

Shenran Wang, Aidan Pine, and Mengzhe Geng. *Exploring Cross-Lingual Voice Conversion Methods for Anonymizing Low-Resource Text-to-Speech.* EACL 2026, Short Papers.

Shenran Wang, Changbing Yang, Mike Parkhill, Chad Quinn, Christopher Hammerly, and Jian Zhu. Developing multilingual speech synthesis system for Ojibwe, Mi'kmaq, and Maliseet. NAACL 2025, Short Papers. **Best Theme Runner-Up Paper Award.**

Research Experience

False Presuppositions in Question Answering 2025.9 – 2026.8

The University of British Columbia | Master's Thesis

- Investigated whether methods for correcting false presuppositions preserve performance on questions with true presuppositions, evaluating across model families, sizes, and benchmarks.
- Identified a tradeoff between correcting false assumptions and answering valid questions; traced overcorrection to weak fact-checking components that reject true presuppositions.
- Designed evaluations and experiments, analyzed model errors, and investigated retrieval-augmented and prompting approaches; led manuscript preparation as first author.

In-Context Learning Across Model Architectures 2025.1 – 2025.7

The University of British Columbia | Undergraduate Research Assistant

- Combined behavioral probing and intervention-based analysis to compare in-context learning in transformer, state-space, and hybrid language models.
- Located function vectors in attention and Mamba layers and showed that their importance depends on whether a task requires parametric knowledge retrieval or contextual understanding. First-author paper in Findings of ACL 2026.

Speech Generation for Indigenous Language Education

Voice Anonymization for Low-Resource Text-to-Speech 2025.6 – 2025.9

National Research Council Canada | Research Intern

- Investigated how voice conversion can mask training speakers' identities while preserving speech naturalness in Plains Cree and SENĆOŦEN text-to-speech systems.
- Compared modifications to pretrained StyleTTS2 models, off-the-shelf voice conversion systems, and cross-lingual speaker transfer. Found that multilingual training with English produced the most consistent results across both languages.
- Designed and evaluated anonymization approaches and led manuscript writing. Highlighted limitations of existing evaluation metrics for cross-lingual voice conversion; first-author paper at EACL 2026.

Efficient Multilingual Speech Synthesis 2024.1 – 2024.9

The University of British Columbia | Undergraduate Research Assistant

- Implemented and evaluated flow-matching multilingual text-to-speech systems for Ojibwe, Mi'kmaq, and Maliseet, studying cross-lingual transfer under limited training data.
- Compared self-attention with attention-free architectures, including FNet, Mamba2, and Hydra. Demonstrated benefits of multilingual training in data-scarce settings and competitive synthesis quality with more memory-efficient architectures.
- Conducted model training, evaluation, and analysis and led manuscript writing. Examined limitations of standard human evaluation for Indigenous language communities; first-author NAACL 2025 paper received the Best Theme Runner-Up Paper Award.

Ongoing Research and Mentorship

Lightweight Real-Time Speech Agents **2026.9 – Present**

The University of British Columbia | Research Project Lead

- Mentored two undergraduate researchers investigating lightweight architectures for real-time speech agents.
- Guided research direction, literature review, and experimental planning.

User Simulation Across Professional Domains **2026.9 – Present**

The University of British Columbia | Research Project Lead

- Gathered and lead a team of graduate students investigating user simulation across different professional domains.
- Developed research ideas and coordinated literature review and project planning.

Additional Experience

SAP | Web Developer Intern, HANA Tooling Team | March – September 2023

- Developed, shipped, and maintained three production features for database management software.
- Volunteered as the first SAP Xi'An internship experience program leader, constructed the intern network at our site and organized events throughout my internship.
- Tech stack includes Kubernetes, Node.js and Java SpringBoot.